

Ex. No. : 10	Overview of Zetta
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Introduction

Zetta is an open-source IoT platform built on Node.js for creating server-based Internet of Things (IoT) applications. It is designed to connect devices, sensors, and cloud services seamlessly through APIs and WebSockets. Zetta enables developers to build real-time, reactive systems that can monitor and control connected devices efficiently.

Key Features

1. Each device connected to Zetta is exposed as a RESTful API, making it easy to interact with devices over the web.
2. Uses the asynchronous and event-driven capabilities of Node.js for fast, scalable IoT applications.
3. Allows devices to publish data streams that can be processed or forwarded to cloud platforms like Azure, AWS, or IBM Bluemix.
4. Runs on small computers like Raspberry Pi, as well as on large cloud servers.
5. Supports real-time communication between devices and applications.
6. Provides easy-to-use tools for developers to simulate, monitor, and manage IoT devices via the Zetta Browser interface.

Architecture

Zetta architecture consists of three main layers:

1. Device Layer:
Physical devices like sensors, actuators, or embedded controllers connected to the system.
2. Zetta Server Layer:
The Zetta server runs on Node.js and acts as a bridge between devices and the Internet.
Each device is represented by a “driver” in Zetta, which defines how the device behaves.
3. Cloud and Application Layer:
Data from devices can be streamed to cloud services or web dashboards for visualization and analytics.

Working Principle

1. Devices are registered in Zetta using drivers.
2. Each driver defines the state and transitions (actions) of the device.
3. The Zetta server exposes devices as RESTful APIs and WebSocket endpoints.
4. Clients (like browsers or mobile apps) communicate with the devices via these APIs.
5. Zetta can connect multiple servers to form a distributed IoT network.

Applications

- Smart home automation
- Environmental monitoring
- Industrial IoT (IIoT) systems
- Connected vehicles
- Remote

device management

Conclusion

Zetta provides a powerful yet flexible platform for building IoT systems that connect physical devices to digital applications.